

Summer School on Rigidity of Discrete Groups, June 30 – July 4, 2025 at IISER Mohali

(Supported by IISER Mohali and The Indian Mathematics Consortium)

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In the proposed meeting, we shall focus on some aspects of the broad topic Rigidity in Geometry and Topology. The geometric form of Mostow’s celebrated rigidity theorem states that every isomorphism between fundamental groups of closed hyperbolic manifolds of dimension at least 3, is induced by an isometry. This result was followed by a similar theorem of Prasad for nonuniform lattices. However, in a spectacular breakthrough, G. A. Margulis obtained a far reaching generalization of these theorems for irreducible lattices in higher rank groups. This result is now known as the Margulis’ Superrigidity Theorem. The ideas behind proofs of these theorems have become very important and fruitful in modern mathematics. In the proposed workshop, we aim to provide a gentle introduction to these results along with similar rigidity theorem in geometry and topology through mini courses and survey talks.

Speakers: Marc Bourdon (U. Lille), Pralay Chatterjee (IMSc), Soma Maity (IISER Mohali), Arghya Mondal (IISER Mohali), C. S. Rajan (Ashoka), Pranab Sardar (IISER Mohali), Riddhi Shah (JNU), T. N. Venkataramana (ICTS Bengaluru).

Expected Participants: Faculty members, PhD students and postdoctoral fellows working in related areas. The topics to be covered in this school require some prior knowledge of geometric group theory, smooth manifolds and Lie groups.

Due to limited funding, we shall be able to provide local hospitality to only a limited number of participants who will be selected based on the proximity of their research interests with the theme of the summer school. Participants are requested to arrange their own travel funding.

Name of the Speaker	No. of Lectures	Abstract
Marc Bourdon (MB), Université de Lille, France	4 lectures	Title: Lie groups and quasi-isometries Abstract: In geometric group theory, one studies groups as geometric objects, and tries to determine whether an algebraic property is a geometric one. In this setting, the natural maps between groups are the so-called quasi-isometries (QI). A class of groups is said to be QI-rigid if every finitely generated group that is QI to a group in the class, is virtually isomorphic to (another) group in the class. Geometric group theory started in the 80's when Gromov proved that the class of nilpotent groups is QI-rigid. In the 90's the subject focused on semisimple groups; it culminated when a combination of works allowed to establish that the class of lattices in a given semisimple group is QI-rigid. Since then, geometric group theory has developed several interactions with other fields of mathematics. It has been used to solve some old open problems (e.g. in 3-manifold topology). A major problem in geometric group theory is to classify connected Lie groups up to QI. One knows that every simply connected Lie group is QI to a completely solvable Lie group, i.e. to a closed subgroup of the upper triangular real matrix group. In 2018 Cornulier conjectured that two QI completely solvable groups must be isomorphic. This is currently open, even in the smaller class of nilpotent Lie groups. During the talks, I plan to introduce some QI-invariants for Lie groups, and illustrate them with examples. These invariants include the growth rate, the rank, and the L^p-cohomology. The examples of groups that will be discussed include the nilpotent groups, the Heintze groups, the abelian-by-abelian solvable Lie groups. Some known results about their QI classification will also be presented.

Name of the Speaker	No. of Lectures	Abstract
Pralay Chatterjee (PC), IMSc Chennai	3 lectures	Title: Introduction to Lattices Abstract: Semisimple Lie groups, Lattices in Lie groups, Some generalities on lattices in Lie groups, $SL_n(\mathbb{Z})$ is a lattice in $SL_n(\mathbb{R})$, Mahler's compactness criteria, Borel density theorem, Arithmetic lattices, Explaining the content of Borel-Harish-Chandra theorem and Margulis Arithmeticity theorem, Godment criteria for co-compactness of arithmetic lattices, Methods to produce arithmetic lattices.
T. N. Venkataramana (TNV) ICTS, Bengaluru	5 lectures	Title: A Proof of the Margulis Superrigidity Theorem Abstract: We shall review the proof of the Margulis superrigidity theorem following the paper by Margulis, Invent. Math. 1984, developing the necessary background. A brief outline is given below. Rank of a semisimple Lie group, Irreducible lattices, Uniform & non-uniform Lattices, Rank 1 vs Higher rank, Examples, Arithmeticity theorem with proof, Zariski dense subgroups, Furstenberg measure, Amenable subgroups, Existence of equivariant measurable maps, Induced representations, Rationality of equivariant measurable maps. Proof of the superrigidity theorem.
C. S. Rajan (CSR), Ashoka University	3 Lectures	Title: Harmonic superrigidity Abstract: The lectures will serve as an introduction to therigidity results of Corlette and Gromov-Schoen.

Name of the Speaker	No. of Lectures	Abstract
Pranab Sardar (PS), IISER Mohali	1 lecture	Title: On some rigidity theorems in geometric group theory Abstract: The rigidity theorems of Mostow, Margulis and others for lattices in semisimple Lie groups inspired many such rigidity type results in geometry and topology. In this expository lecture, we will survey some such theorems related to geometric group theory.
Arghya Mondal	2 lectures	Title: Mostow rigidity Abstract: The Mostow Prasad Rigidity Theorem says if two finite volume hyperbolic manifolds of dimension greater than or equal to 3 are homotopy equivalent then they are isometric. We will give a complete proof of this statement for closed hyperbolic 3-manifolds. Then we will indicate how the proof can be extended in the general case.
Riddhi Shah (RS)	1 lecture	Title: The structure of automorphism groups and lattices in Lie groups Abstract: We study the structure of groups of automorphisms of a connected Lie group; namely, we identify certain conditions under which they are almost algebraic, and discuss some applications and examples (joint work with S.G. Dani - https://arxiv.org/abs/2504.18641). We explore the structure of lattices in a connected Lie group and discuss some properties of automorphisms which keep a lattice invariant (joint work with Rajdip Palit and Manoj B. Prajapati - Geometry, and Dynamics 17 (2023), 185-213 - https://doi.org/10.4171/GGD/672)

references

- [Dave Witte Morris - Arithmetic groups](#)
- [Venkataramana-rigidity-notes.pdf](#)
- [Raghunathan - Discrete subgroups of Lie groups](#)
- [Marc Bourdon \(Université de Lille\): "Quasi-isometric inv of continuous group \$L^p\$ cohomology & apps."](#) [Marc Bourdon - Quasi-isometric invariants of continuous group \$L^p\$ cohomology and applications](#)
- [Robert J. Zimmer - Ergodic Theory and Semisimple Groups \(1984, Birkhäuser\).pdf](#)

- [Werner Ballmann, Mikhael Gromov, Viktor Schroeder - Manifolds of Nonpositive Curvature \(1985, Birkhäuser\).pdf](#)

What is a rigidity theorem?

Mostowls rigidity theorem

Quasi-isometric rigidity

Let X, Y are metric spaces and $k \geq 0$. Then

$$f : X \rightarrow Y$$

is called a **k -quasi isometric embedding** if

$$-k + \frac{1}{k}d(x, x') \leq d(f(x), f(x')) \leq k + d(x, x')$$

for all $x, x' \in X$.

Examples

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$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R}^2 \\ t &\mapsto (t, \cos t) \end{aligned}$$

is a quasi embedd

-

$$t \mapsto (t, t^2)$$

is **not** a quasi embedding

-

$$t \mapsto (t, |t|)$$

is a quasi-isometry onto image



$$\mathbb{R}^m \cong_{\text{quasi}} \mathbb{R}^n \implies n = m$$



$$H^n \cong_{\text{quasi}} H^m \implies n = m$$

All our spaces are either manifolds... or graphs!

Consider a connected graph X and assign length 1 to each edge, which makes X a metric space

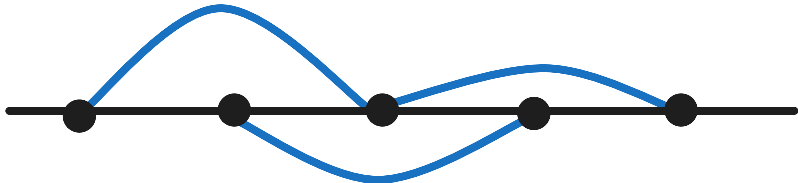
$$d(x, y) := \inf\{\text{length of paths from } x \text{ to } y\}$$

Now given a group Γ and $S \subset \Gamma$ we may consider its Caley graph $\text{Cay}(\Gamma, S)$ with vertices Γ and edges on (g, gs) for all $g \in \Gamma$ and $s \in S$.

Exercise

$\text{Cay}(\Gamma, S)$ is connected if $\Gamma = \langle S \rangle$.

Consider $\Gamma = \mathbb{Z}$ and $S = \{1, 2\}$ then the Caley graph is



(Milnor-Schwarz) Let X be a length metric space and $\Gamma \curvearrowright X$ acts by isometries properly such that X/Γ is compact then Γ is finitely generated and

$$\begin{aligned} \Gamma &\rightarrow X \\ g &\mapsto gx \end{aligned}$$

Corollaries:

- $\Gamma_1 < \Gamma$ (f.g.) and Γ/Γ_1 is finite then

$$\Gamma_1 \hookrightarrow \Gamma$$

is a quasi-isometry

- Let Γ be f.g. and $F \trianglelefteq \Gamma$ with $|F| < \infty$ then

$$\Gamma \rightarrow \Gamma/F$$

is a quasi-isometry.

- $$\text{Cay}(\Gamma, S_1) \cong_{\text{quasi}} \text{Cay}(\Gamma, S_2)$$
 if $\langle S_1 \rangle = \langle S_2 \rangle = \Gamma$

 Let X is a Riemannian symmetric space and $\Gamma < \text{Isom}(X)$ is a uniform lattice. Then

$$\begin{aligned} \Gamma &\rightarrow X \\ g &\mapsto gx \end{aligned}$$

is a quasi isometry.

Examples:

- $\mathbb{Z}^n \curvearrowright \mathbb{R}^n$
- $\pi_1(S_g) \curvearrowright H^2, g \geq 2$ then $\Gamma \cong_{\text{quasi}} H^2$

 **Definition. Virtual isomorphism of groups**

Suppose Γ_1, Γ_2 are two f.g. groups. We say that Γ_1, Γ_2 are **virtually isomorphic** if there exists $\Gamma'_1 < \Gamma_1, \Gamma'_2 < \Gamma_2$ finite index and $F_1 \trianglelefteq \Gamma'_1, F_2 \trianglelefteq \Gamma'_2$ finite such that

$$\frac{\Gamma'_1}{F_1} \cong \frac{\Gamma'_2}{F_2}$$

 **Intuition**

Suppose $\mathfrak{p}$ is some group theoretic property and Γ is a group. We say that Γ satisfies $\mathfrak{p}$ **virtually** if Γ has a finite index subgroup satisfying $\mathfrak{p}$.

For example being *virtually abelian*.

 **Definition. Quasi-rigidity**

A group Γ is called **quasi-rigid** if $\Gamma \cong_{\text{quasi}} \Gamma' \implies$ is virtually isomorphic to Γ'

 **Intuition**

A group theoretic property $\mathfrak{p}$ is called **quasi-rigid** if $\Gamma_1 \cong_{\text{quasi}} \Gamma_2$ and Γ_1 has property $\mathfrak{p}$ implies there exists a Γ_3 with property $\mathfrak{p}$ that is virtually isomorphic to Γ_2 .

 **(Gromov's polynomial growth theorem, Milnor, Wolf)** If Γ is quasi-isometric to a nilpotent group then Γ is virtually nilpotent.

 **(Stallings, Dunwoody et al)** If $\Gamma \cong_{\text{quasi}} F_n, n \geq 2$ then Γ is virtually free.

Example:

- $SL(2, \mathbb{Z}) = \mathbb{Z}_4 *_{\mathbb{Z}_2} \mathbb{Z}_6$ is virtually free.

 **Keywords**

ends of groups, ends of metric spaces, chain of unbohnded components, throw balls, ...

 **(Gromov)** Γ is quasi-isometric to $\mathbb{Z}^n$ implies Γ is virtually isomorphic to $\mathbb{Z}^n$.

 $\Gamma \cong_{\text{quasi}} \mathbb{Z}$ implies Γ is virtually cyclic

The first step is:

Proposition: Γ is Gromov hyperbolic.

and then use:

 Γ infinite Gromov hyperbolic $\implies \Gamma$ has an infinite order element.

Let $\gamma \in \Gamma$ be an infinite order element.

The second step is:

 Let H hyperbolic group and $x \in H$ has infinite order. Then $\langle x \rangle \hookrightarrow H$ is a quasi-isometric embedding.

Hence $\mathbb{Z} \cong \langle \gamma \rangle \hookrightarrow_{\text{q-emb}} \Gamma \rightarrow_{\text{qi}} \mathbb{Z}$

- $\implies$ the composition is a qi

This implies

 **Example**

$\langle \gamma \rangle \hookrightarrow \Gamma$ is a qi

This implies

Proposition: $\Gamma / \langle \gamma \rangle$ is finite

proved by the fact that image of a quasi-isometry is coarsely surjective:

 **Exercise**

If $f : X \rightarrow Y$ is a quasi-isometry then there exists $D \geq 0$ such that for all $y \in X \exists x \in X$ such that
$$d(f(x), y) \leq D$$

 **(Dyabina?)** Solvability is **not** a quasi-rigid property.

Examples of quasi-rigid properties:

- finite presentation
- Amenability
- hyperbolicity

- virtual cohomological dimension

Examples of quasi-rigid groups (quasi isometric implies virtually isomorphic):

- MCG of $S_g, g \geq 2$
- $\pi_1(S_g), g \geq 2$ (Gabai, Tukia, Cassen et al)
- π_1 of closed hyperbolic manifolds of $\dim \geq 3$ are quasi-rigid (Tukia, Sullivan)
- (Richard Schwartz) π_1 of finite volume hyperbolic ≥ 3 -manifolds are quasi-rigid
- Suppose X is an irreducible symmetric space of non-compact type of rank ≥ 2

-  **(Klienier, Leeb)** Uniform lattices in $\text{Isom}(X)$ are quasi-rigid.

-  **(Eskin)** Non-uniform lattices of $\text{Isom}(X)$ are quasi-rigid.

A non-example:

- (Burger-Mozhes) $F_2 \times F_2$ is not quasi-rigid


Problems

- Are polycyclic groups quasi-rigid.
- Are $SL(2, \mathbb{Z}) \times SL(n, \mathbb{Z}), n \geq 3$ quasi-rigid
- Are hyperbolic groups the form F_n


Keywords

max cohomological dim of MCG

Pralay Chatterjee - Introduction to Lattices


Outline

- Generalities on lattices in Lie groups
- $SL(n, \mathbb{Z})$ is a lattice in $SL(n, \mathbb{R})$
- Borel density theorem
- state Borel-Harish-Chandra theorem
- Godment criteria for co-compactness of arithmetic lattices
- state Margulis Arithmeticity theorem for lattices in higher rank Lie groups
- Methods to produce arithmetic lattices

All groups will be second countable and locally compact. But mostly they will be Lie groups.


Every locally compact group G posses a left Haar measure and it is unique up to a multiplicative constant.


Definition. Modular function of a locally compact group

Let μ be a left invariant Haar measure. And

$$\begin{aligned} \mathfrak{B}_G &\rightarrow \mathbb{R} \\ A &\mapsto \mu(Ag) \end{aligned}$$

Then we have the modular function

$$\Delta : G \rightarrow \mathbb{R}_{>0}$$

$$\mu(Ag) := \Delta(g^{-1})\mu(A)$$

or all $A \in \mathfrak{B}_G$.

It is a continuous group homomorphism.

If G is a Lie group then we can choose a top form $\omega \in \Omega^{\dim G}(G)$. This gives a left invariant Haar measure.

Then it is easy to calculate the modular function:

Let $\mathfrak{g}$ be the Lie algebra of G . Then we have the adjoint representation

$$\begin{aligned} \text{Ad} : G &\rightarrow GL(\mathfrak{g}) \\ \mathfrak{g} &\mapsto \text{d}_i c_g \end{aligned}$$

and

$$\Delta(g) = |\det \text{Ad}(g)|$$

 **Definition.** If $\Delta(G) = \{1\}$ then G is called **unimodular**.

Following groups are unimodular:

- compact groups
- G such that $G = [G, G]$
- semi-simple G
- connected nilpotent Lie group

However,

- solvable groups are **not** in general unimodular

All our measures are *always* regular.

 **Definition.** ρ -semi invariant measures

Let

- $\rho : G \rightarrow \mathbb{R}_{>0}$ be a continuous character
- $(X, \mathfrak{M}, \mu)$ be a measure space
- $G \curvearrowright X$ be measurable group action

then μ is said to be **ρ -semi invariant** if $\mu(gA) = \rho(g)\mu(A)$ for all $g \in G, A \in \mathfrak{M}$.

Let $H < G$ be a closed subgroup of a Lie group G .

Then G/H is a manifold and

$$G \curvearrowright G/H$$

 Let $H < G$ be a closed subgroup of a Lie group(?or locally compact?) G . Let $\rho : G \rightarrow \mathbb{R}_{>0}$ be a continuous character. Then $\exists$ ρ -semi invariant measure on G/H

$$\Longleftrightarrow$$

$\rho(h) = \Delta_G(h)\Delta_H(h^{-1})$ for all $h \in H$. Moreover, (given fixed ρ) if one of the above holds, any two such measures will be positive multiples of each other.

Reference

- (*corollary*) If H is unimodular G/H admits a Δ -semi invariant measure
- (*corollary*) If $\Gamma < G$ is discrete, then G/H admits a Δ_G -semi invariant measure

Proposition: Suppose G is a Lie group and $H < G$ and H is unimodular. Suppose G/H and

- (*corollary*) If H is unimodular G/H admits a Δ -semi invariant measure

where μ is such a measure such that $\mu(G/H) < \infty$ then G is unimodular.



$$\mu\left(g\frac{G}{H}\right) = \Delta(g)\mu\left(\frac{G}{H}\right)$$

thus $\Delta(g) = 1$

📌 **Definition.** Let G be a Lie group and $\Gamma < G$ is a discrete subgroup then Γ is a **lattice** if G/Γ admits a G -invariant measure such that $\mu(G/\Gamma) < \infty$.

📌 **Definition.** Let $\Gamma < G$ is discrete. Then a Borel set $F \subset G$ is called a **fundamental domain** for Γ if $G = F\Gamma$ and $F \cap F\gamma = \emptyset$ if $\gamma \neq i_G$.

📖 Let $\Gamma < G$ is any discrete subgroup then it admits a fundamental domain .

💡 Let $U \subset G$ be open such that $U^{-1}U \cap \Gamma = \{i_G\}$

- G is second countable $\implies$

$$Ug_nU = G$$

-

$$F := \bigcup_{n=1}^{\infty} \left(g_n U - \bigcup_{i < n} g_i U \Gamma \right)$$

Proposition: If F is a fundamental domain, $\gamma_i \in \Gamma$ and $F = \bigcup_{i \geq 1} A_i$ disjoint. Then

$$\bigcup_{i \geq 1} A_i \gamma_i$$

is another fundamental domain. Any fundamental domain can be obtained from another by such a process.

Proposition: If $C \subset G$ such that

$$\frac{C}{G} = \frac{G}{\Gamma} \iff C\Gamma = G$$

then there exists a fundamental domain $F \subset C$.

Proposition: If $B \subset G$ is such that $B \rightarrow \frac{G}{\Gamma}$ is injective then there exists a fundamental domain $F \supset B$.

📌 **Definition.** Let ν be the right invariant Haar measure on G and F be a fundamental domain for Γ . Then

$$\nu_{G/\Gamma}(A) := \nu_G(\pi^{-1}(A) \cap F)$$

for all Borel $A \subset G/\Gamma$ and $\pi : G \rightarrow G/\Gamma$.

 Exercise

Check that $v_{G/\Gamma}$ is independent of the fundamental domain chosen.

 Use

Proposition: If F is a fundamental domain, $\gamma_i \in \Gamma$ and $F = \bigcup_{i \geq 1} A_i$ disjoint. Then

$$\bigcup_{i \geq 1} A_i \gamma_i$$

is another fundamental domain. Any fundamental domain can be obtained from another by such a process.

This measure is

- $B \subset G$ then $\nu_G(B) \geq \nu_{G/\Gamma}(\pi(B))$ and equal when π is injective on B
- regular*: for compact $K \subset G/\Gamma$ choose $C \subset G$ compact such that $\pi(C) = K$ then

$$\infty > \nu_G(C) \geq v_{G/\Gamma}(K)$$

- Δ -semi invariant



1.12. Theorem. *Let G be a locally compact group satisfying the second axiom of countability and $\Gamma \subset G$ a lattice. Let $\{x_n\}_{1 \leq n < \infty}$ be a sequence in G . Let $\pi: G \rightarrow G/\Gamma$ be the natural map. Then the sequence $\pi(x_n)$ has no convergent subsequence in G/Γ if and only if there exists a sequence $\{\gamma_n\}_{1 \leq n < \infty}$ in Γ such that $\gamma_n \neq e$ for any n and $x_n \gamma_n x_n^{-1}$ converges to e as n tends to ∞ .*

Riddhi Shah - The structure of automorphism groups and lattices in Lie groups

 Abstract

We study the structure of groups of automorphisms of a connected Lie group; namely, we identify certain conditions under which they are almost algebraic, and discuss some applications and examples (joint work with S.G. Dani - <https://arxiv.org/abs/2504.18641>). We explore the structure of lattices in a connected Lie group and discuss some properties of automorphisms which keep a lattice invariant (joint work with Rajdip Palit and Manoj B. Prajapati - Geometry, and Dynamics 17 (2023), 185-213 - <https://doi.org/10.4171/GGD/672>)

$$\mathrm{Aut}(G) \rightarrow \mathrm{AutLieAlg}(\mathfrak{g}) \subset GL(\mathfrak{g})$$

is one to one

$$\exp(d\tau(X)) = \tau(\exp X)$$

 **Definition.** $\mathrm{Aut}(G)$ is **almost algebraic** if it is open subgroup finite index in an algebraic group.

 **(Wigner, 1976)** $\mathrm{Aut}(G)_0$ is almost algebraic.

 **(Dani 1992)** Let C be the max cpt connected central (max central tori) subgroup of G . Then $\mathrm{Aut}(G)$ is almost algebraic if C is trivial.

	Aut
T^n	$GL(n, \mathbb{Z})$

	Aut
$\mathbb{R}^n$	$GL(n, \mathbb{R})?$
simply connected G	AutLieAlg($\mathfrak{g}$)? Zariski closed? algebraic...
G/C as in (Dani 1992) Let C be the max cpt connected central (max central tori) subgroup of G. Then $\text{Aut}(G)$ is almost algebraic if C is trivial.	$\text{Aut}(G/C)$ is almost algebraic
N_3	
$SL(2, \mathbb{R}) \backslash \text{semi} G$	almost alg
$SO(2, \mathbb{R}) \backslash \text{semi } G$	not alg!

....